MILITARY SPECIFICATION

LOAD BOX, ELECTRICAL: T-4165

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Load Box Electrical: T-4165 which is used on the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8857859	- Load Box, Electrical: T-4165.
ET-8857859	- Equipment Tabulation.
8857861	- Panel Assembly, Load Box Unit B.
8857880	- Cable Assembly, Special Purpose, Electrical: T-4165-1.
8857894	- Panel Assembly, Load Box Unit A.
8858048	- Cable Assembly, Special Purpose, Electrical: T-4165-2.
8860998	- Box Packing, Ammunition, For Load Box, Electrical, T-4165, Atomic Ordnance Test Equipment.

FSC: 1190

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The electrical load box shall comply with all requirements specified on the drawing (dwg) 8857859, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Circuit Resistance.

3.5.1 Unit A. The resistance between Load Box A panel terminals and electrical connector J3 terminals listed in Table I shall be as specified when tested in accordance with 4.4.1.

Table I

Circuit resistance, load box unit A

From (Panel Terminals)	To (Conn. J3 Terminal)	Resistance (Ohm)
E1	A	0.10 max
E2	D	0.10 max
E3	B	0.10 max
E4	D	0.10 max
E5	C	0.10 max
E6	D	0.10 max
E5	L (b)	9.1K max
E5	L (a)	13.5K ± 6%
E7	E8 (d)	1.5 max
E7	E8 (e)	100 ± 10%
E8	S (j) (k)	0.10 max
E8	J (i)	0.10 max
E9	M	0.10 max
E10	D	0.10 max

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E11	P	82 ± 12%
E12	D	0.10 max
E13	N	0.10 max
J3-R	E7 (c)	0.10 max
E14	D	0.10 max
E15	F	2.2K ± 11%
E16	D	0.10 max
E17	E	2.2K ± 11%
E18	D	0.10 max
E19	L	0.10 max
E20	K (f)	0.10 max
E21	J	0.10 max
E22	H	0.10 max
E23	J	0.10 max
E24	G	0.10 max
Conn. J3-N	D (f)	0.15 max
Conn. J3-R	J (g)	0.10 max
Conn. J3-K	L (h)	0.10 max
Conn. J3-T	J (i)	0.10 max
Conn. J3-T	J (j)	0.10 max
E10	E11	82 ± 12%

(a) With 6A1A1 ADJUST knob (R1) in extreme CW position.
(b) With 6A1A1 ADJUST knob (R1) in extreme CCW position.
(c) With switch S3 in position A and switch S2 in position B.
(d) With switch S3 in pos. OFF and 6A1A5 ADJUST knob (R2) in extreme CW position.
(e) With switch S3 in pos. OFF and 6A1A5 ADJUST knob (R2) in extreme CCW position.
(f) With switch S1 in position B.
(g) With switch S2 in position A.
(h) With switch S1 in position A.
(i) With switch S3 in position A.
(j) With switch S3 in position B.
(k) With 6A1A5 ADJUST knob (R2) in extreme CW position.

3.5.2 Unit B. The resistance between Load Box Unit B panel terminals listed in Table II shall be as specified when tested in accordance with 4.4.1.

Table II

Circuit resistance, load box unit B

From Terminal	To Terminal	Resistance (Ohm)
E1	E4	1.8 ± 0.02 K
E6	E5	330 ± 3.3
E9	E12	12 ± 0.12 K
E13	E16	350 ± 3.5
E17	E20	110 ± 1.1
E21	E24	49 ± 0.49
E25	E28	470 ± 4.7

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E29	E32	8 ± 0.08
E35	E38 (a)	1.8 ± 0.02 K
E35	E38 (b)	1.75 ± 0.02 K
E41	E42 (c)	0.645 ± 0.025
E41	E42 (d)	0.675 ± 0.035
E39	E42 (e)	Open
E39	E41	0.05 max
J2-G	E41	0.05 max
J2-H	E41	0.05 max
J2-h	E41	0.05 max
J2-f	E42	0.05 max
J2-x	E42	0.05 max
J2-Y	E42	0.05 max
J2-KK	E42	0.05 max
J2-JJ	E39	0.05 max
J2-Z	E31 (b) (f)	0.10 max
J2-Y	E31 (a)	0.10 max
J2-w	E38	0.10 max
J2-A	E37	0.10 max
J2-n	E34	0.10 max
J2-m	E33	0.10 max
J2-u	E32	0.10 max
J2-GG	E32	0.10 max
J2-NN	E31	0.10 max
J2-MM	E31	0.10 max
J2-q	E28	0.10 max
J2-U	E27	0.10 max
J2-p	E24	0.10 max
J2-T	E23	0.10 max
J2-L	E20	0.10 max
J2-i	E19	0.10 max
J2-F	E16	0.10 max
J2-g	E15	0.10 max
J2-e	E12	0.10 max
J2-C	E11	0.10 max
J2-AA	E8	0.10 max
J2-z	E5	0.10 max
J2-j	E3	0.10 max
J2-s	J1C	0.10 max
J2-s	J2r	0.10 max
J2-W	J1A (g)	0.10 max
J2-V	J1A (g)	0.10 max
J2-W	J1A (h)	Open
J2-V	J1A (h)	Open
J1-8	Ground Lug	0.10 max
E35	E36	0.10 max
E29	E30	0.10 max
E25	E26	0.10 max
E21	E22	0.10 max
E17	E18	0.10 max
E13	E14	0.10 max

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E9	E10	0.10 max
E6	E7	0.10 max
E4	E5	0.10 max
E1	E2	0.10 max

With switches S2 thru S10 in OFF position:

E2	E3	Open
E7	E8	Open
E10	E11	Open
E14	E15	Open
E18	E19	Open
E22	E23	Open
E26	E27	Open
E30	E31	Open
E36	E37	Open

With switches S2 thru S10 in ON position:

E2	E3	0.10 max
E7	E8	0.10 max
E10	E11	0.10 max
E14	E15	0.10 max
E18	E19	0.10 max
E22	E23	0.10 max
E26	E27	0.10 max
E30	E31	0.10 max
E36	E37	0.10 max

(a) With switch S12 in 330 V position.
(b) With switch S12 in 380 V position.
(c) With switch S12 in 6.0 V position and S11 in ON position.
(d) With switch S12 in 7.0 V position and S11 in ON position.
(e) With switch S11 in OFF position.
(f) With switch S9 in OFF position.
(g) With switch S1 in ON position.
(h) With switch S1 in OFF position.

3.6 Insulation resistance. The insulation between each terminal and all other terminals listed for the respective assemblies in Tables III and IV shall be greater than 20 megohms when tested at 500 + 50 Direct Current Volts (vdc) as specified in 4.4.2. Unused terminals shall be connected together and tested as one terminal. The panel of each assembly shall be at test fixture ground.

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Table III

Load Box Unit A Connector J3 Terminals

(For this test, switches S1 and S2 shall be in position B, switch S3 shall be in the OFF position.)

J3 - A, B, C, D, E, F, G, H, J, K, M, R, S, T, U.
Panel Terminals: E7 (6A1A5 Input + Red)
J3 Unused Terminals: V, W, X, Y, Z, a, b, c.

Table IV

Load Box Unit B Connector Terminals

(For this test all toggle switches shall be in the OFF position. Range switch S12 shall be in the 6.0 volt position.)
J1 - A, C.
J2 - A, C, F, G, K, L, T, U, V, Z, a, c, f, g, i, j, k, m, n, p, q, u, v, w, z, AA, BB, HH, MM, PP.

J2 Unused Terminals: B, D, E, J, N, P, R, S, X, b, c, d, t, CC, DD, EE, FF.

3.7 Environmental testing. The electrical load box shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The electrical load box shall also comply with 3.5 and 3.6 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.7.1 Transportation vibration. The electrical load box assembled in packing box, dwg. 8860998, shall withstand vibration in the vertical plane (vertical plane is defined as that plane in which the assembly is to be shipped) at the frequencies and amplitudes specified in Table V without evidence of damage or affecting function when tested as specified in 4.4.3.1. Upon completion of the vibration, the electrical load box shall meet the requirements of 3.5 and 3.6. In addition, the special purpose cable assembly and the electrical power cable assembly shall meet the requirements for circuit resistance and insulation resistance as specified on dwgs. 8857880 and 8858048, respectively.

Table V

Frequency (Hz)	Sweep Time Minutes	Double Amplitude Inches	Number of Sweeps
5-10-5	10 ± 0.5	0.04 ± 10%	1
11-15-11	10 ± 0.5	0.25 ± 10%	1
16-20-16	10 ± 0.5	0.13 ± 10%	1
21-60-21	15 ± 0.5	0.06 ± 10%	1

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3.8 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8857859) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

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Table VI

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Circuit resistance (see 3.5)		X
3. Insulation resistance (see 3.6)		X
4. Transportation vibration (see 3.7.1)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VII and Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Load box unit A (see dwg. 8857894).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Component loose or missing	Visual/Manual
	102. Solder joint defective	Visual/Manual
	103. Crimp incomplete	Visual/Manual
	104. Objective evidence relating to material improper or missing (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.9)	Visual
	202. Component markings illegible or obliterated	Visual
	203. Panel markings illegible or obliterated	Visual
	204. Serialization improper or missing	Visual

4.3.2.1.2 Load box unit B (see dwg. 8857861).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Component loose or missing	Visual/Manual
	102. Solder joint defective	Visual/Manual

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103. Crimp incomplete.....Visual/Manual
104. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.9).....Visual
202. Component markings illegible or obliterated.....Visual
203. Panel markings illegible or obliterated.....Visual
204. Serialization improper or missing (see 3.4).....Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
	101. Preservation packaging and packing not in accordance with applicable drawing	Visual
	102. Marking not in accordance with applicable drawing	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.9)	Visual

4.3.3 Testing. All tests shall be performed. Tests performed in the order specified below:

Test	Test classification
a. Circuit resistance	Major
b. Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance and insulation resistance. Each load box shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.5 and 3.6, the item shall be rejected and removed from the lot.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). Els for assemblies are listed on the Equipment Tabulation, ET-8857859. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified onthe EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The continuity and resistance shall be measured between the terminals listed in 3.5.1 and 3.5.2 with a four-wire resistance measuring instrument.

4.4.2 Insulation resistance. Insulation resistance shall be measured with an insulation resistance tester at 500 ± 50 vdc with current limited to 5 milliamperes. For this test, all unused terminals of the connector shall be connected together and tested as one terminal. The panel of each assembly shall be at test fixture ground.

4.4.3 Environmental testing (see 3.7).

4.4.3.1 Transportation vibration. Place the packing box containing the electrical load box on a suitable vibration machine and vibrate in the vertical plane for the durations, frequencies, and amplitudes specified in Table V. At the completion of vibrations, the electrical load box shall be inspected for evidence of damage and tested for the requirements of 3.5 and 3.6. The Special Purpose Cable, T-4165-1, and the Electrical Power Cable, T-4165-2, shall be inspected for circuit resistance and insulation resistance as specified on dwgs. 8857880 and 8858048, respectively.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The load box shall be packaged in accordance with dwg 8860998 using the specified moisture barrier.

5.1.2 Level C. The load box shall be packaged in accordance with dwg 8860998 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The load box shall be packed in accordance with dwg 8860998.

5.3 Marking. Marking shall be as specified in accordance with dwg 8860998.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2266 Rev. I dated 4 January 1966.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A079

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